MIL-C-915/66A
1 August 1972
SUPERSEDING
MIL-C-915/66(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 59-CONDUCTOR, TYPE MSPW

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

Conductor size 22-7 (16 required):

(A) First - Copper conductor, ASTM B-286 size 22-7, tin coated.
Second - Fluorocarbon resin insulation, nominal thickness 0.044-inch. (Taped insulation may be used.)
Third - Braided shield of AWG No. 34 or No. 36 copper. Minimum coverage 85 percent; braid angle 30 to 35 degrees.
Fourth - Shield insulation consisting of one polyester tape, type G of MIL-I-631, opaque white, with a jacket of transparent polyamide, type III, grade E or type IV of MIL-M-20693 colored yellow. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Inner tape opaque white; outer tape yellow.

For either type of shield insulation, use Standard Identification Code, applied by Method 2, on the inner tape, using numbers 30 through 45.

Conductor size 20-7 (18 required):

(A) First - Copper conductor, ASTM B-286 size 20-7, tin coated.
(A) Second - Polyvinyl chloride insulation (105°C.), colored white. Nominal thickness 0.022 inch.
(A) Third - Insulation jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.
Fourth - Braided shield of AWG No. 34 or No. 36 copper, tin coated. Minimum coverage of 85 percent; braid angle of 30 to 35 degrees.
(A) Fifth - Shield insulation consisting of one polyester tape, type G of MIL-I-631, with a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed.

For either type of shield insulation, use Standard Identification code, applied by Method 2, on the inner tape, using numbers 12 through 29.

Sixth - Conductors numbered 12, 13 and 14 formed into a triad. Nominal lay 3 inches.
Seventh - Conductors numbered 19 through 24 formed into three pairs, with the following pairing: Numbers 19 and 20; numbers 21 and 22; numbers 23 and 24. Nominal lay 3 inches.

(A) Denotes changes

FSC 6145

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Conductor size 3(7) (25 required):

First - Copper conductor, Navy Standard size 3(7), tin coated.
Second - Polyvinyl chloride insulation (105° C.). Nominal wall thickness 0.014 inch. Eleven conductors colored white; eleven conductors colored black; three conductors colored red.
(A) Third - Jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693, over each conductor. Nominal thickness 0.003 inch.
Fourth - Sixteen conductors formed into eight pairs, each pair consisting of one white and one black conductor. Nominal lay 3 inches.
Fifth - Nine conductors formed into three triads, each triad consisting of one white, one red and one black conductor. Nominal lay 3-1/2 inches.
Sixth - Over each pair and each triad, a braided shield of AWG No. 34 or No. 36 copper, tin coated. Minimum coverage 85 percent; braid angle of 30 to 35 degrees.
(A) Seventh - Shield insulation, over each shield, consisting of one polyester tape, type G of MIL-I-631, with a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed.

For either type of insulation, use Standard Identification Code, applied by Method 2, on the inner tape. The triads shall be numbered 1, 2 and 3; the pairs shall be numbered 4 through 11.

Assembly:

First - A solid polychloroprene filler as the center of the core. Non-fibrous fillers shall be employed throughout the cable.
Second - Inner layer, consisting of the following:
(1) The shielded triads numbered 1, 2 and 3.
(2) The triad consisting of the conductors numbered 12, 13 and 14.
(3) The conductors numbered 15, 16, 17 and 18.
Third - A binder tape.
Fourth - Middle layer, consisting of the following:
(1) The eight shielded pairs numbered 4 through 11.
(2) The three pairs consisting of conductors numbered 19 and 20, 21 and 22, 23 and 24.
Fifth - A binder tape.
Sixth - Outer layer, consisting of the following:
(1) The conductors numbered 25 through 45, cabled consecutively.
Seventh - A binder tape.
(A) Eighth - Polychloroprene jacket (black) nominal thickness 0.110 inch. Cable overall diameter 1.565 inches minimum, 1.635 inches maximum. Cable surface marking required.

EXAMINATION AND TESTS:	Requirements
Basic Electrical:	
Voltage withstand - Volts rms, minimum	
Size 22 conductors	
Conductor to conductor	5000
Conductor to shield	3000
Shield to shield	200
Sizes 20 and 3(7) conductors	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	200
Insulation resistance - Megohms/1000 ft., minimum	
Size 22 conductors	5000
Size 20 and 3(7) conductors	150
Conductor resistance - At 25°C.; ohms/1000 feet, maximum	
(A) Size 22 conductors	17.71
(A) Size 20 conductors	11.03
(A) Size 3(7) conductors	4.23

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	Requirements:
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Characteristic impedance (size 22 conductors only)	
- At 1 MHz; ohms	55 ± 5
Capacitance (size 22 conductors only)	
- At 1 kHz; pF/ft., maximum	27
Group B:	
Drip - At 75° + 1°C.	Zero
Cold bending cable - At 0° + 2°C.; over 18-inch maximum diameter mandrel.	No damage
Physicals (unaged)	
Insulation (polyvinyl chloride only)	
Tensile strength - psi, minimum	1800
Elongation - percent, minimum	100
Cable jacket	
Tensile strength - psi, minimum	1800
Elongation - percent, minimum	300
Set - Inch, maximum	30/64
Group C:	
Flammability - Inches, maximum	2
Physicals (aged)	
Insulation (polyvinyl chloride only)	
Air oven	
Elongation - Percent of unaged, minimum	90
Cable jacket	
Air pressure heat	
Tensile strength - Percent of unaged, minimum.	65
Elongation - Percent of unaged, minimum.	65
Hot oil immersion	
Tensile strength - Percent of unaged, minimum.	65
Elongation - Percent of unaged, minimum.	65
Permanence of printing (jacket) - Cycles, minimum.	250
Cable filler removability	

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cable aging and compatibility (95° ± 3°C.)	No failure

UNIT ORDERING LENGTHS: 500 feet (nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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